

Pillar 5: Capstone Project

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Post Graduate Diploma in Artificial Intelligence and Machine Learning

PHASE 07 | FetalGuard AI (CTG — Fetal Risk Classifier) — GenAI Helper Interface

GenAI Helper Interface Documentation

Overview

A Generative AI interface was added to improve usability, interpretation, and reporting. The interface does not replace the predictive model; instead, it provides natural-language support for exploratory analysis and prediction explanation. It helps users understand the dataset, interpret model outputs, and obtain simple recommendations from the system.

Inputs

The interface accepts two main types of input. First, users can upload a CSV file containing the dataset. This file is used for exploratory analysis and data dictionary generation. Second, users can enter patient-level features in JSON format for prediction explanation. Each JSON key should match a model feature name, and each value should represent the corresponding measurement for one case.

How the Input Works

When a CSV file is uploaded, the app loads it into a tabular structure and extracts summary statistics such as row count, column count, missing values, numeric distributions, and categorical frequencies. These statistics are passed to the LLM, which converts them into a concise narrative summary. For the data dictionary, the app collects column names, data types, missing counts, unique values, and sample entries, then sends this metadata to the LLM for documentation generation.

When JSON features are entered, the app uses them as the direct input to the prediction explanation workflow. The JSON object represents one record of patient data. The model

prediction value, such as 0 for normal or 1 for abnormal, is then interpreted by the LLM together with the feature values and confidence score.

How the Prediction Works

The predictive model analyzes the uploaded feature values and outputs a class label. This label is not the final explanation; it is only the model's decision. The LLM then receives the prediction, confidence, and feature information and produces a human-readable interpretation. This design separates numerical prediction from textual explanation, making the system more transparent and easier to use.

Step-by-Step Use

1. Open the Streamlit application.
2. Upload a CSV dataset file.
3. Review the preview table shown in the app.
4. Click Generate EDA Summary to obtain a plain-English dataset overview.
5. Click Generate Data Dictionary to generate column documentation.
6. Enter patient features in valid JSON format.
7. Select the predicted class label.
8. Adjust the confidence score if available.
9. Click Explain Prediction to receive a natural-language explanation.
10. Read the recommended next action if the app provides one.

Role of Generative AI

Generative AI is used in three ways:

- to summarize dataset characteristics,
- to document features automatically,
- to explain model predictions in language that non-technical users can understand.

This improves accessibility and supports communication between technical and non-technical stakeholders.

Conclusion The GenAI interface extends the machine learning workflow by adding interpretability and automation. It enables faster understanding of data, clearer documentation, and more intuitive prediction explanations. The result is a more practical and user-friendly application for analysis and decision support.